

Kryptos K4 and the ELYOIE Jigsaw Matrix

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Yet another speculative idea for Kryptos K4

In the present document, I'm exploring the idea that Kryptos K4 uses a Quagmire III cipher with primary key word KRYPTOS and a running secondary key that's assembled, in **JIGSAW** puzzle fashion, from a unique matrix of 98 characters. The latter is derived from Kryptos K3 plaintext via simple steps, including transposition and substitution with key (surprise!) KRYPTOS.

This matrix, which I'd like to call the ELYOIE Jigsaw Matrix, has a perfectly positioned match of key **ELYOIE** with cipher text **NYPVTT**, yielding the known plaintext of **BERLIN**. From the matrix, we're also able to recover the key to the known plaintext of **CLOCK**, **EAST** and **NORTHEAST**.

Finally, there are indications of K4 plaintext starting with the word **JIGSAW**, possibly continuing with the words **HIDDEN ON CIA YARD** and ending with the word **SKILLS**, as emphasized by **color** and demonstrated in the table below.

Kryptos K4 cipher text	? OBKR UOXOGHULBSOLIFBBWFLRVQQPRNGKSS ○ TWTQSSJQSSEKZZWATJKLUDIAWINFB NYP VTTMZFPKWGDKZXTJCDIGKUHUA UEKCAR
Super-imposed plaintext, both known plaintext and proposed plaintext	? JIGS AWHIDDENONCIAYARDEASTNORTHEAST ○ TWTQSSJQSSEKZZWATJKLUDIAWINFB BER LINCLOCKWGDKZXTJCDIGKUHUA SKILLS
Target Matrix: Required Quagmire III secondary key, with respect to primary key KRYPTOS	? INGU HBDJPTDXPGVYBDRAGRDUMRIYWOYNKY ○ TWTQSSJQSSEKZZWATJKLUDIAWINFB ELY OIECBAQKWGDKZXTJCDIGKUHUA IEEMJU
ELYOIE Jigsaw Matrix	INGUH WBMWIKVYBGTPSXMSRJDMAWIEEMYEYH OHRDVCYVRDGACLBGELEGRHSASNDEELY OIEKGMADJPTCBAQSUQCTTBMJXTEFCMU

Interestingly, the private K1 work sheet that Jim Sanborn released in 2010 [1] already contains a check-marked hint that K4 plaintext might start with the very word **JIGSAW**, but more on that later. Presently, you'll see a step-by-step description of how to derive the ELYOIE Jigsaw Matrix.

Deriving the ELYOIE Jigsaw Matrix

This is how to build the ELYOIE Jigsaw Matrix.

Initial input

Take K3's plaintext. Three things stick out: The misspelled word DESPARATLY and the solitary characters X and Q.

SLOWLY DESPARATLY SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT
ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED WITH
TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND
CORNER AND THEN WIDENING THE HOLE A LITTLE I INSERTED THE
CANDLE AND PEERED IN THE HOT AIR ESCAPING FROM THE CHAMBER
CAUSED THE FLAME TO FLICKER BUT PRESENTLY DETAILS OF THE ROOM
WITHIN EMERGED FROM THE MIST X CAN YOU SEE ANYTHING Q

Resolve anomalies

Let's resolve the anomalies of K3 plaintext. First, we correct the misspelled word DESPARATLY. Then, we discard the extraneous X after the word MIST, and finally, we elide the solitary Q at the end. Thus, we arrive at the following text of 335 characters:

SLOWLY DESPERATELY SLOWLY THE REMAINS OF PASSAGE DEBRIS THAT
ENCUMBERED THE LOWER PART OF THE DOORWAY WAS REMOVED WITH
TREMBLING HANDS I MADE A TINY BREACH IN THE UPPER LEFT HAND
CORNER AND THEN WIDENING THE HOLE A LITTLE I INSERTED THE
CANDLE AND PEERED IN THE HOT AIR ESCAPING FROM THE CHAMBER
CAUSED THE FLAME TO FLICKER BUT PRESENTLY DETAILS OF THE ROOM
WITHIN EMERGED FROM THE MIST X CAN YOU SEE ANYTHING Q

Arrange into 8x42 matrix

We now arrange the text into a matrix of 8 lines and 42 columns. The matrix is one character short from being filled completely, but this is no cause for concern.

SLOWLYD	ESPERAT	ELYSLOW	LYTHERE	MAINSOF	PASSAGE
DEBRIST	HATENCU	MBEREDT	HELOWER	PARTOFT	HEDOORW
AYWASRE	MOVEDWI	THTREMB	LINGHAN	DSIMADE	ATINYBR
EACHINT	HEUPPER	LEFTHAN	DCORNER	ANDTHEN	WIDENIN
GTHEHOL	EALITTL	EIINSER	TEDTHEC	ANDLEAN	DPEERED
INTHEHO	TAIRESC	APINGFR	OMTHECH	AMBERCA	USEDTHE
FLAMETO	FLICKER	BUTPRES	ENTLYDE	TAILSOF	THEROOM
WITHINE	MERGEDF	ROMTHEM	ISTCANY	OUSEEAN	YTHING

Next, it's time for transposition and substitution steps with key word KRYPTOS, as explained in the following.

Block transposition with key word KRYPTOS

Let's now apply a block transposition with key word KRYPTOS:

SYWLDLO EAESTRP EOSLWLY LRHYEET MONAFSI PGSAEAS
DSRETIB HCEAUNT MDRBTEE HEOERWL PFTATOR HROEWOD
ARAYESW MWEIDV TMRHBET LAGINHN DDMSEAI ABNTRYI
ENHATIC HEPERPU LATENHF DERCRNO AETNNHD WIEINND
GOETLHH ETIALTL EENIRSI TETECHD AALNNED DEEPDRE
IHHNOET TSRACEI AFNPRGI OCHMHET ACEMARB UHDSETE
FTMLOEA FECLRKI BEPUSRT EDLNEYT TOLAFSI TORHMOE
WNHIEIT MDGEFER RETOMHM INCSYAT OAEUNES YGITNH

Line-based substitution with key word KRYPTOS

Line by line, we do alternating encryption/decryption steps as per the table below. All substitutions are done with the Quagmire III cipher, primary key word KRYPTOS.

The secondary key, which in this step is only 1 character long, changes with each line as per the following table:

line #	1	2	3	4	5	6	7	8
operation	encrypt	decrypt	encrypt	decrypt	encrypt	decrypt	encrypt	decrypt
secondary key	K	R	Y	P	T	O	S	K

Please note that the first and the last lines remain unchanged. This is because the character 'K' happens to be the initial character of our Quagmire III primary key, KRYPTOS.

SYWLDLO EAESTRP EOSLWLY LRHYEET MONAFSI PGSAEAS
COKDPHA GBDSQMP LCKAPDD GDTDKVJ YEPSPTK GKTDVTC
CPCTGBZ QZGALFX SQPJGDS NCILUJU FFQBGCL CDUSPTL
BJETRFS EBKBXKM HTRBJEC ABXSXJY TBRJJEA QFBFJJA
LCIBUMM IBNEUBU IIWNODN BIBIGMH EEUWWIH HIIAHOI
DCCHKSZ ZRVYTSZ YAHXVBD KTCGCSZ YTSGYVP JCORSZS
MDXWELG MLIWASU HLCRFAD LJWZLBD DEWGMFU DEAQXEL
WNHIEIT MDGEFER RETOMHM INCSYAT OAEUNES YGITNH

Note: The Crypto Crap blog [3] features a very noteworthy screenshot of, and a very shrewd comment on, the K3 matrix worksheet that Jim Sanborn showed on NOVA, which was marked with alternating P and C characters in front of the rows. The line-based alternating substitution step is inspired by that valuable content.

Arrange into 67x5 matrix

Re-arrange the text into a matrix of 67 lines and 5 columns. The key **ELYOIE** is already crawling down the last column.

SYWLD
LOEAE
STRP**E**
OSLW**L**
YLRH**Y**
EETM**O**
NAFS**I**
PGSA**E**
...
UNESY
GITNH

Transpose into 5x67 matrix

Transpose the 67x5 matrix into a 5x67 matrix.

SLSOYENPADBPPTYTDPZLPNJBDLRKHESBAJIIIBNIHWIDSYADCSJZWLFWFDFQNTFTIAUG
YOTSLEAGSPDLDDDEKVCQFJCUGUBFBTCXRQJBBOBEIACZTHKSGCSEIHAZEUXHMEONTNI
WERLRTFSCHSCDKPGTTZXDIFCSJSXRAJJFAUNIDIEHHCZSXTZYOMLWLDLWDEIDRMCOET
LAPWHMSAOAQKGVSKCGSGLFLPEEKBBYJBLME**INGUH**OHRDVCYVRDGACLBGELEGRHSASN
DE**ELYOIE**KGMADJPTCBAQSUQCTTBMJXTEFCMUWBMWIIKVYBGTPSXMSRJDMAWIEEMYEH

Cut off last 98 characters

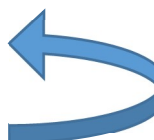
Kryptos K4 consists of a question mark that's followed by 97 Latin characters, for a total of 98 characters. Let's reduce the 5x67 matrix by keeping only the last 98 characters, matching the length of Kryptos K4.

INGUHOHRDVCYVRDGACLBGELEGRHSASNDE**ELYOIE**KGMADJPTCBAQSUQCTTBMJXTEFCMU
WBMWIIKVYBGTPSXMSRJDMAWIEEMYEH

Lay out same as Kryptos K4, then apply finishing touch

Arrange the remaining 98 characters in the same way as the unique grid of Kryptos K4. Then, as a finishing touch, push the last line to a position just after the first line. This happens to settle the key **ELYOIE** at the exact position where Kryptos K4 has got the fitting cipher text **NYPVTT** to the plaintext of **BERLIN**.

INGUH
OHRDVCYVRDGACLBGELEGRHSASNDE**ELY**
OIEKGMADJPTCBAQSUQCTTBMJXTEFCMU
WBMWIIKVYBGTPSXMSRJDMAW**IE**EMYEH



We're done. The result is the ELYOIE Jigsaw Matrix:

INGUH
WBMWIIKVYBGTPSXMSRJDMAW**IE**EMYEH
OHRDVCYVRDGACLBGELEGRHSASNDE**ELY**
OIEKGMADJPTCBAQSUQCTTBMJXTEFCMU

Starting the Jigsaw Puzzle

In November of 2010, when Jim Sanborn announced the **BERLIN** clue [2], Sanborn also released his own private work sheet [1] of K1 showing a check mark that he himself had made under the artefact KE-ING. I interpret this mark as a hint that the key (KEI) to K4 starts with the three characters **ING**:

...D**ING**A...

...S**E**STP...

...J**K**ZLD...

✓

Taking the check-marked key (KEI) of **ING** and feeding it into a Quagmire III cipher with primary key KRYPTOS, the first three characters **OBK** of K4 decrypt to **JIG**. There are not many reasonable ways in which to continue this trigram, the most obvious being the word **JIGSAW**.

The **ELYOIE** Jigsaw Matrix presented herein, which has the same length as K4, starts off with the very KEI characters **ING** that are check-marked in Jim Sanborn's private decryption chart, and the plaintext proposed in the present document indeed starts with the word **JIGSAW**.

Jim Sanborn's K1 chart also contains, as various people have noted, a cleverly hidden - and similarly check-marked - hint to the key **ELYOIE** to **BERLIN**.

Disclaimer

The present document is a decryption attempt only. Besides, it's also a work in progress. Please note that anything, absolutely everything that's presented here may be attributed to chance and happenstance alone. Like so many other decryption attempts, the one presented here is likely to come to a dead end while filling the remaining gaps. It's all about trial and error, you know...

Note

An earlier version of the present document was called "Kryptos K4 and the ELYOIE Match Matrix". The matrix specified herein is 100% identical to the one shown in the earlier document.

Final note

Dear fellow Kryptos fans, I'd like to thank you for reading this document!

[1] <https://www.nytimes.com/2010/11/21/us/21codecharts.html>

[2] <https://www.nytimes.com/2010/11/21/us/21code.html>

[3] <http://cryptocrap.blogspot.com/2011/07/kryptos.html>

Annex: Characteristics of ELYOIE Jigsaw Matrix

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Super-imposed plaintext, both known plaintext and proposed plaintext	? JIGS AWHIDDENONCIAYARDEASTNORTHEAST ○ TWTQSSJQSSEKZZWATJKLUDIAWINFB BER LINCLOCK WGDZKXTJCDIGKUHUA SKILLS
Target Matrix: Required Quagmire III secondary key, with respect to primary key KRYPTOS	? INGU HBDJPTDXPGVYBDRAGRDUMRIYWOYNKY ○ TWTQSSJQSSEKZZWATJKLUDIAWINFB ELY OIECBAQK WGDZKXTJCDIGKUHUA IEEMJU
ELYOIE Jigsaw Matrix	INGUH WB MWI IKVYBG T P SXMS R DM AW IEEMY Y H O HRDVC Y VRDGA CLBGELEG R HSAS NDE ELY OIEK GMAD DJPTCBAQ S U QCTTBM JXTEFCMU
Exact-position matches are shown with grey background	

Characteristics of key matches:

Known Plaintext	Cipher text	Key	Contiguous key length	Location of key vs. location of cipher text
BERLIN	NYPVTT	ELYOIE	complete – 6 out of 6	perfect match – ELYOIE key exactly aligned with NYPVTT
CLOCK	MZFPK	CBAQK	4 + 1 out of 5	same line
EAST	FLRV	RDUM	jigsaw puzzle	exact-position match of R , M and final Y
NORTHEAST	QQPRNGKSS	RIYWOYNKY		

Proposed plaintext	Cipher text	Key	Contiguous key length	Location of key vs. location of cipher text
JIGSAW	OBKRUO	INGUHB	5 + 1 out of 6	off by one: just skipping the '?'
HIDDEN	XOGHUL	DJPTDX	4 + 1 + 1 out of 6	hidden
ON	BS	PG	N/A	same line
CIA	OLI	VYB	3 out of 3	same line, off by three
YARD	FBBW	DRAG	scrambled	next line
SKILLS	UEKCAR	IEEMJU	4 + 1 + 1 out of 6	exact-position match of last key character U